Enterprise Network with Dual-ISP Redundancy: Technologies Overview

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1. Project overview

This project simulates the network of a multi-site enterprise with redundant Internet access. It was built and tested in an PNETLab environment with Cisco IOL routers and switches.

Main objectives

- Interconnect the enterprise sites with dynamic routing.
- Give the whole enterprise Internet access through two ISPs (Ooredoo and Algérie Télécom), with automatic failover.
- Provide gateway redundancy for the user networks with VLANs and HSRP.
- Hide the private addressing with PAT.
- Keep the configuration simple: the Internet default route is configured on a single edge router and distributed to all other routers by OSPF.

2. Technologies used

Technology	Purpose
OSPF (multi-area)	Dynamic routing inside the enterprise, with a backbone area and additional areas
RIP v2	Dynamic routing for the part of the network that does not run OSPF
Route redistribution	Exchange of routes between OSPF and RIP
Static default routes	Internet access through the two ISPs
Floating static route	Backup default route with a higher administrative distance
IP SLA and tracking	Detects an ISP failure beyond the interface and triggers the failover
`default-information originate`	Distributes the default route to all OSPF routers
NAT/PAT with route-maps	Translates private addresses to the address of the active ISP link
VLANs and 802.1Q trunks	Separates user groups on shared switches
Router-on-a-stick	Inter-VLAN routing with sub-interfaces
HSRP per VLAN	Gateway redundancy and load sharing between two routers

3. Routing design

3.1 OSPF (multi-area)

- **Area 0** is the backbone. It contains the core routers and the links between them.
- Additional areas connect to the backbone through **ABRs**, routers with one interface in each area.
- Areas reduce the size of the link-state database and the CPU load, and they contain the effect of a link failure inside one area.
- Interfaces facing the users are declared **passive**, so no OSPF hellos are sent toward them.

3.2 RIP and redistribution

- The part of the network that does not run OSPF uses **RIP v2**.
- A boundary router (an **ASBR**) runs OSPF and RIP together and redistributes between them

3.3 Default route to the Internet

Only the edge router has static default routes:

- **Primary route:** through the main ISP, with the default administrative distance, installed only while the tracking object is Up.
- **Backup route:** through the second ISP, a floating static route with a higher administrative distance.
- **Host route for the probe:** a /32 route to the IP SLA target through the main ISP only, so the probe always tests the primary link.

The edge router then runs `default-information originate` under OSPF. All other OSPF routers learn the default route automatically as `O*E2 0.0.0.0/0`. No default route is configured on them.

4. Internet redundancy: IP SLA and tracking

An interface can stay up while the ISP behind it is broken, so a static route alone would not fail over. IP SLA solves this.

1. The edge router sends an ICMP echo every 10 seconds to a target in the Internet, **through the primary ISP link**.
2. A tracking object is Up when the probe gets replies and Down when it does not. A delay (down 10, up 20) prevents flapping.
3. When the probe fails, the primary default route is removed from the routing table and the floating route through the second ISP becomes active.
4. When the probe works again, the primary route returns (preemption).

Result: the failover happens automatically in about 15 seconds, and OSPF keeps distributing a default route during the switch.

5. NAT/PAT

Private networks (172.16.0.0/16, 192.168.0.0/16 and 10.0.0.0/8) cannot be routed on the Internet, so the edge router translates them.

Without PAT, a ping from a user network reaches the Internet but the reply cannot come back, because the Internet has no route to private networks.

6. LAN redundancy: VLANs and HSRP per VLAN

6.1 VLANs and trunks

- Two VLANs are created on every switch that uses them.
- Links between switches and toward the gateway routers are **802.1Q trunks**. User ports are **access ports** in a single VLAN.
- The gateway routers use **sub-interfaces** (one per VLAN) for inter-VLAN routing.

6.2 HSRP per VLAN

Two routers share a virtual gateway address per VLAN, and the roles are split to share the load:

	VLAN A	VLAN B
Router 1	**Active** (priority 110)	Standby (priority 100)
Router 2	Standby (priority 100)	**Active** (priority 110)

- **Preempt** lets a router recover its role after a failure.
- **Interface tracking** on the uplink lowers the priority by 20 if that link fails, so the other router takes over.
- Both routers run **HSRP version 2**. Mixing versions prevents them from seeing each other.
- The PCs use the **virtual IP** as their gateway, never the real router address.

Result: if one router fails, its VLAN moves to the other router and users keep their connectivity without any change on the PCs.

7. Traffic flow example

A user PC pings a destination on the Internet:

1. The PC sends the packet to its virtual gateway, answered by the HSRP Active router.
2. The gateway router forwards it toward the core using the OSPF default route.
3. The core forwards it to the edge router, which originates the default route.
4. The edge router sends it through the primary ISP and translates the source to the address of that link.
5. The ISP forwards it to the Internet, and the reply returns to the translated address.
6. The edge router translates the reply back and returns it to the PC along the same chain.

If the primary ISP fails, only step 4 changes: the packet leaves through the second ISP, with the source address of that link.

8. Verification commands

Goal	Command
OSPF neighbors	`show ip ospf neighbor`
OSPF routes and default route	`show ip route ospf`
RIP status	`show ip protocols`, `show ip route rip`
IP SLA probe	`show ip sla statistics`
Tracking object	`show track`
Active default route	`show ip route static`
NAT translations	`show ip nat translations`
HSRP roles	`show standby brief`
VLANs and trunks	`show vlan brief`, `show interfaces trunk`